

WorkPilot AI — Tech Stack & Architecture Report

1. Technologies Currently Used in the Project

- **Frontend Framework:** React 18, TypeScript 5, Vite 6, Tailwind CSS 4, Framer Motion, Lucide Icons
- **State & Context API:** AuthContext, AppContext, WorkPilotChatContext, GmailContext, GoogleCalendarContext, LlmContext
- **AI Engine:** builtInChatbot.ts (50 Context-Aware Capabilities, Priority Evidence Calculator, Agentic Actions)
- **Backend Proxies:** Vite Node IMAP Dev Server Proxy (vite/gmailImapPlugin.ts), Google iCal Racing Engine
- **PDF & Reporting:** Executive Workday & Tech Stack Print PDF Engine (workdayPdf.ts, techStackPdf.ts)
- **Persistence:** HTML5 LocalStorage Caching & Hydration Layer

2. Future Technologies Planned (Roadmap)

- **Backend Services:** Python FastAPI / Node.js NestJS microservices + PostgreSQL (Prisma ORM) + Redis Cache
- **Vector DB & RAG:** Qdrant / Pgvector / Pinecone for Document & Policy Embeddings
- **Production LLMs:** OpenAI GPT-4o, Google Gemini 1.5 Pro, Anthropic Claude 3.5 Sonnet, DeepSeek-R1 (Ollama)
- **Voice AI & Sync:** WebSockets / Socket.io + OpenAI Whisper / Web Speech API
- **Cloud & CI/CD:** Netlify / Vercel Edge CDN, Docker Containers, GitHub Actions CI/CD